

# C SAPP.c (修改完善)

⇒ 其中所有函数均可重入 (reentrant), 不依赖全局/静态可变状态.

私有辅助函数.

C code

```
static size_t str_strlen(const char s[]) {  
    int i = 0;  
    while (s[i] != '\0') ++i;  
    return i;  
}
```

1. 反转字符串

C code

```
static void str_reverse(char s[]) {  
    int i, j;  
    for (i = 0, j = str_strlen(s) - 1; i < j; i++, j--) {  
        char c = s[i];  
        s[i] = s[j];  
        s[j] = c; }  
}
```

2. 将十进制转换为b进制的字符串

C code

```
static void str_itoa(long u, char s[], int k) {  
    # itoa K&R 异读  
    # inter to ASCII  
    int i = 0, j;  
    int neg = u < 0;  
    if (neg) u = -u;  
    do {  
        c = (u % k);  
        s[i++] = (c < 10) ? (c + '0') : (c - 10 + 'a'); # 利用 '0' 作为基准, 以余数作为偏移  
    } while (u /= k > 0); # 局限: 只能将数反转, 以 'a' 作偏移输出16进制.  
    if (neg) s[i++] = '-';  
    j = i;  
    s[i] = '\0';  
    int start = 0, end = j - 1;  
    while (start < end) {  
        char tmp = s[start];  
        s[start] = s[end];  
        s[end] = tmp;  
        start++;  
        end--; }  
}
```